

A Single Stepsize Suffices for Unprojected Linear TD(0)

Simultaneous Robust and Fast Rates via Polyak–Ruppert Averaging

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Setting I: from RL successes to policy evaluation



Reinforcement learning learns by interaction.

- **State** s_t : what the agent sees now.
- **Action** a_t : what the agent does next.
- **Policy** μ : a rule mapping states to actions.
- **Reward** r_t : the immediate feedback received after taking an action.

AlphaGo's success

- Once a policy is fixed, we want to know quickly: *how good is this policy from each state?*

Policy evaluation

For a fixed policy μ , estimate

$$V^\mu(s) = \mathbb{E} \left[\sum_{k \geq 0} \gamma^k r_k \mid s_0 = s \right]$$

the expected discounted return from state s with discount factor $\gamma \in (0, 1]$.

Setting II: Linear TD(0) with Markovian sampling

In large state spaces, exact tabular values are too expensive. Use linear approximation:

$$V_{\boldsymbol{\theta}}(s) = \phi(s)^\top \boldsymbol{\theta}, \quad \phi(s) \in \mathbb{R}^d, \quad \|\phi(s)\| \leq \phi_\infty.$$

Let $Z_t = (s_t, s_{t+1})$, unprojected linear TD(0) updates by the TD direction:

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t + \eta_t \mathbf{g}(\boldsymbol{\theta}_t, Z_t),$$

$$\mathbf{g}(\boldsymbol{\theta}, Z_t) = (r_t + \gamma \phi(s_{t+1})^\top \boldsymbol{\theta} - \phi(s_t)^\top \boldsymbol{\theta}) \phi(s_t).$$

The stationary TD direction and TD fixed point are

$$\bar{\mathbf{g}}(\boldsymbol{\theta}) := \mathbb{E}_{Z \sim \pi_Z}[\mathbf{g}(\boldsymbol{\theta}, Z)], \quad \boldsymbol{\theta}^* : \quad \bar{\mathbf{g}}(\boldsymbol{\theta}^*) = \mathbf{0}.$$

Performance measure: the TD potential

For $v \in \mathbb{R}^{|S|}$, define the stationary value norm and Dirichlet seminorm

$$\|v\|_D^2 := \sum_s \pi(s) v(s)^2, \quad \|v\|_{\text{Dir}}^2 := \frac{1}{2} \sum_{s,s'} \pi(s) P^\mu(s, s') (v(s) - v(s'))^2.$$

We measure error by the TD potential

$$f(\theta) = (1 - \gamma) \|\mathbf{V}_\theta - \mathbf{V}_{\theta^*}\|_D^2 + \gamma \|\mathbf{V}_\theta - \mathbf{V}_{\theta^*}\|_{\text{Dir}}^2.$$

Intuition

The potential rewards two things at once:

- $\|\mathbf{V}_\theta - \mathbf{V}_{\theta^*}\|_D$: value accuracy under the stationary distribution
- $\|\mathbf{V}_\theta - \mathbf{V}_{\theta^*}\|_{\text{Dir}}$: consistency along Markov transitions.

Performance measure: two finite-time regimes

The potential has curvature around the TD fixed point θ^* :

$$f(\theta) - f(\theta^*) \geq \omega \|\theta - \theta^*\|^2, \quad \omega := (1 - \gamma)\lambda_{\min}(\Phi^\top D \Phi) > 0.$$

Two regimes of interest

Robust rate

works even when curvature is tiny

$$\tilde{O}(1/\sqrt{T})$$

Fast rate

exploits curvature when it helps

$$\tilde{O}(1/(\omega T))$$

Robust when ω is small; fast when ω is large.

Why stability is hard I: Markovian sampling

The sample $Z_t = (s_t, s_{t+1})$ is generated by the same trajectory that produced θ_t . Thus, in general,

$$\mathbb{E}[g(\theta_t, Z_t) \mid \mathcal{F}_{t-1}] \neq \bar{g}(\theta_t).$$

Why this matters

- For i.i.d. data, the update noise is centered after conditioning on the past.
- For Markov data, the current sample still remembers the previous state, so the noise has a **bias** that must be controlled.

Why stability is hard II: self-amplification and prior fixes

Self-amplification.

$$\|g(\theta_t, Z_t)\| \leq r_\infty \phi_\infty + 2\phi_\infty^2 \|\theta_t\|.$$

A loose bound on $\|\theta_t\|$ makes the bound on the update $\|g(\theta_t, Z_t)\|$ loose, which feeds back into θ_{t+1} .

What prior work uses

- **Projection:** (Bhandari et al. 2018; Liu and Olshevsky 2021). Stable by design, but changes TD and requires a projection radius.
- **Curvature / contractive stability:** (Srikant and Ying 2019; Patil et al. 2023; Samsonov et al. 2024; Mitra 2025; Chandak and Borkar 2025; Durmus et al. 2025; Li et al. 2026). Projection-free or lightly modified, but rate, burn-in, or tuning can depend on curvature.

Question: one plain TD stepsize for both regimes?

Question

Can unprojected TD(0) achieve high-probability stability and simultaneous robust and fast rates without knowledge of ω ?

Best of both worlds target

$$f(\bar{\theta}_T) - f(\theta^*) \leq \tilde{O}\left(\min\left\{\frac{1}{\sqrt{T}}, \frac{1}{\omega T}\right\}\right)$$

- If ω is small: still keep a $1/\sqrt{T}$ guarantee.
- If ω is large: automatically obtain the fast $1/(\omega T)$ rate.

Main result I: one ω -agnostic stepsize gives stability

Use a single decaying stepsize

$$\eta_t = \frac{1}{c \tau_{\text{mix}} \phi_\infty^2} \frac{1}{\sqrt{t+1} \log(t+3)}.$$

Let

$$R_{\text{base}} = \max \left\{ \|\theta_0 - \theta^*\|, \|\theta^*\|, \frac{r_\infty}{\phi_\infty} \right\}.$$

Mixing assumption

Assume the Markov chain $Z_t = (s_t, s_{t+1})$ is irreducible and aperiodic with stationary law π_Z . Then,

$$\max_{z \in \mathcal{Z}} \left\| P_Z^k(z, \cdot) - \pi_Z \right\|_{\text{TV}} \leq 4 \cdot 2^{-k/\tau_{\text{mix}}}, \quad k \geq 0.$$

Implicit boundedness

For sufficiently large c , with probability at least $1 - \delta$,

$$\sup_{t \geq 0} \|\theta_t\| \leq R_{\text{max}}.$$

No projection is applied to the iterates.

Main result II: simultaneous robust and fast rates

Define the weighted Polyak–Ruppert average

$$S_T = \sum_{t=1}^T \eta_{t-1}, \quad \bar{\boldsymbol{\theta}}_T = \frac{1}{S_T} \sum_{t=1}^T \eta_{t-1} \boldsymbol{\theta}_{t-1}.$$

Main theorem

For a sufficiently large c depending on δ , set $R_{\max} = \rho(c, \delta) R_{\text{base}}$. With probability at least $1 - \delta$, for all $T \geq 1$,

$$f(\bar{\boldsymbol{\theta}}_T) - f(\boldsymbol{\theta}^*) \leq \tilde{\mathcal{O}}\left(R_{\max}^2 \min\left\{\frac{\tau_{\text{mix}}^2 \phi_{\infty}^4}{\omega T}, \frac{\tau_{\text{mix}} \phi_{\infty}^2}{\sqrt{T}}\right\}\right).$$

Stability proof I: make the loop self-bounding

Induct on the past event

$$\text{IH}_{t-1} := \left\{ \max_{0 \leq i < t} \|\boldsymbol{\theta}_i\| \leq R_{\max} \right\}.$$

Since $\|\boldsymbol{\theta}_t\| \leq \|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*\| + \|\boldsymbol{\theta}^*\|$, it is enough to control $\|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*\|$.

Expanding the squared distance gives

$$\begin{aligned} \|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*\|^2 &= \|\boldsymbol{\theta}_0 - \boldsymbol{\theta}^*\|^2 + \sum_{i=0}^{t-1} \eta_i^2 \|\mathbf{g}_i\|^2 \\ &\quad + 2 \sum_{i=0}^{t-1} \eta_i \langle \bar{\mathbf{g}}(\boldsymbol{\theta}_i), \boldsymbol{\theta}_i - \boldsymbol{\theta}^* \rangle + 2 \sum_{i=0}^{t-1} \eta_i h_i(Z_i), \end{aligned}$$

where

$$h_i(z) := \langle \mathbf{g}(\boldsymbol{\theta}_i, z) - \bar{\mathbf{g}}(\boldsymbol{\theta}_i), \boldsymbol{\theta}_i - \boldsymbol{\theta}^* \rangle.$$

Stability proof I: isolate the only hard term

The mean drift helps:

$$\langle \bar{g}(\theta_i), \theta_i - \theta^* \rangle = -[f(\theta_i) - f(\theta^*)] \leq 0.$$

Under IH_{t-1} , the quadratic term is controlled because $\sum_i \eta_i^2 < \infty$.

Only hard term: Markov bias

$$B_t := \sum_{i=0}^{t-1} \eta_i h_i(Z_i).$$

This is not a standard martingale sum because Z_i is Markovian and h_i changes with θ_i .

Stability proof II: Poissonize the Markov bias

Fix θ_i . Since $\mathbb{E}_{Z \sim \pi_Z}[h_i(Z)] = 0$, solve the Poisson equation

$$u_i - P_Z u_i = h_i, \quad u_i(z) = \sum_{\ell=0}^{\infty} (P_Z^\ell h_i)(z), \quad \|u_i\|_\infty \leq 16\tau_{\text{mix}} \|h_i\|_\infty.$$

Under IH_{t-1} ,

$$\|h_i\|_\infty \lesssim \phi_\infty^2 R_{\text{max}}^2, \quad \|u_i\|_\infty \lesssim \tau_{\text{mix}} \phi_\infty^2 R_{\text{max}}^2.$$

Each Markov-noise term decomposes as

$$h_i(Z_i) = \underbrace{u_i(Z_{i+1}) - (P_Z u_i)(Z_i)}_{\text{martingale difference}} + \underbrace{u_i(Z_i) - u_i(Z_{i+1})}_{\text{telescoping remainder}}.$$

Two key implications

- The first term is controlled by Pinelis' inequality from bounded differences.
- The second term is rewritten by Abel summation; the moving u_i 's are controlled by Lipschitz continuity and $\theta_i - \theta_{i-1} = \eta_{i-1} g_{i-1}$.

Together, these bounds make B_t small enough to feed back into the bootstrap.

Stability proof III: close the bootstrap

The previous bounds give

$$\|\theta_t\|^2 \leq \left(4 + \frac{A_1(\delta)\rho^2}{c} + \frac{A_2\rho^2}{c^2}\right) R_{\text{base}}^2.$$

To close the induction, it is enough that

$$4 + \frac{A_1(\delta)\rho^2}{c} + \frac{A_2\rho^2}{c^2} \leq \rho^2.$$

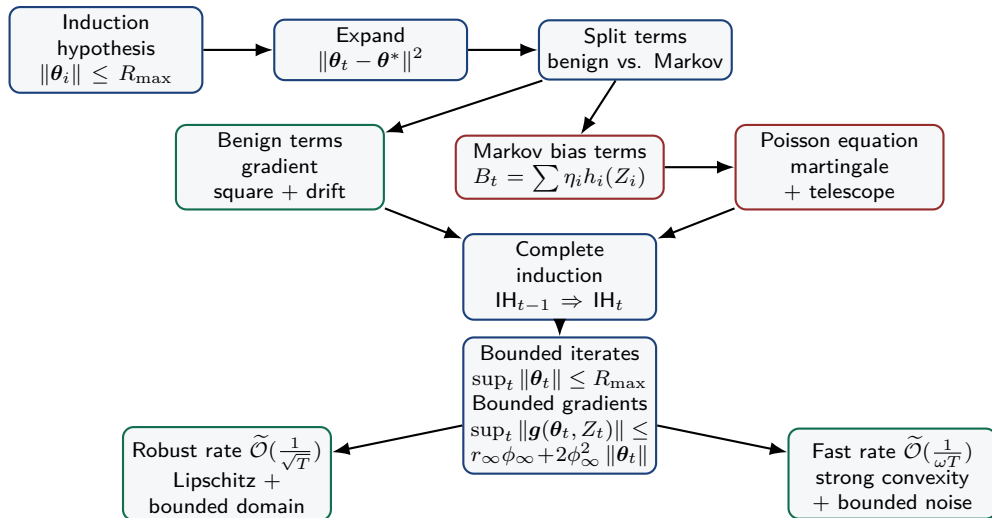
Many pairs (c, ρ) close the induction

- As $c \rightarrow \infty$, the admissible radius factor satisfies $\rho \downarrow 2$.
- There is a minimum lower bound $c > c_{\min}(\delta)$.
- Larger c : smaller stepsize, smaller radius bound.
- Smaller c above the threshold: larger stepsize, larger radius bound.

Bootstrap conclusion

$$\text{IH}_{t-1} \implies \text{IH}_t, \quad \sup_{t \geq 0} \|\theta_t\| \leq \rho R_{\text{base}}.$$

Proof road map: stability first, rates second



Conclusion

Main message

A single ω -agnostic stepsize suffices for projection-free linear TD(0) under Markov sampling.

- **Algorithmic takeaway:** no projection radius and no tuning by the curvature/eigenvalue parameter ω .
- **Statistical takeaway:** the averaged iterate gets simultaneous high-probability rates

$$\min \left\{ \tilde{\mathcal{O}}\left(\frac{1}{\omega T}\right), \tilde{\mathcal{O}}\left(\frac{1}{\sqrt{T}}\right) \right\}.$$

- **Proof takeaway:** self-bounding induction gives implicit boundedness; a Poisson-equation decomposition controls Markov bias.

Open problem: Can we still get a reasonable best-of-both-worlds rate using a τ_{mix} -agnostic stepsize?

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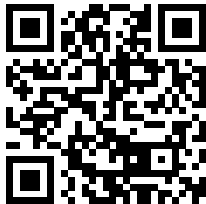
Open problem: Can we still get a reasonable best-of-both-worlds rate using a τ_{mix} -agnostic stepsize? (Currently, $\eta_t = \frac{1}{c \tau_{\text{mix}} \phi_{\infty}^2} \frac{1}{\sqrt{t+1} \log(t+3)}$, but τ_{mix} is usually unknown in practice.)

Thank you

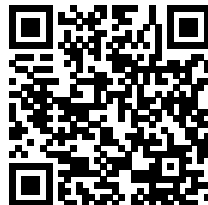
Thank you! Any questions?



Slides



Paper



Personal website

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